```
586                 writer = csv.DictWriter(f, fieldnames=headers)
587                 writer.writeheader()
588                 for row in rows:
589                     writer.writerow(row)
590         except PermissionError:
591             # Common on Windows when CSV is open in Excel; do not break API/logging.
592             return None
593 
594         return str(out_path)
595 
596     def _append_trajectory_csv_rows(self, rows):
597         if not rows:
598             return
599         day_key = datetime.now(timezone.utc).strftime("%Y%m%d")
600         csv_path = self.append_csv_dir / f"trajectory_{day_key}.csv"
601         headers = [
602             "flightKey",
603             "timestamp",
604             "icaoAddress",
605             "callsign",
606             "latDD",
607             "lonDD",
608             "altitudeM",
609             "speedCMS",
610             "headingDE2",
611             "verVelocityCMS",
612             "squawk",
613             "sourceIp",
614         ]
615         write_header = not csv_path.exists() or csv_path.stat().st_size == 0
616         with csv_path.open("a", newline="", encoding="utf-8") as f:
617             writer = csv.DictWriter(f, fieldnames=headers)
618             if write_header:
619                 writer.writeheader()
620             for row in rows:
621                 writer.writerow(row)
622 
623     def _purge_old_history_if_needed(self):
624         if self.history_retention_days <= 0:
625             return
626 
627         today = datetime.now(timezone.utc).date()
628         if self._last_purge_day == today:
629             return
630 
631         cutoff = datetime.now(timezone.utc) - timedelta(days=self.history_retention_days)
632         cutoff_iso = cutoff.isoformat()
633         cur = self._conn.cursor()
634         cur.execute("DELETE FROM trajectory_samples WHERE timestamp < ?", (cutoff_iso,))
635         self._conn.commit()
636         self._last_purge_day = today
637 
638     def get_trajectory_history(self, flight_key=None, hours=24, limit=200000):
639         if not self.enabled:
640             return []
641 
642         limit = max(1, int(limit))
643         hours = max(1, int(hours))
644         cutoff_iso = (datetime.now(timezone.utc) - timedelta(hours=hours)).isoformat()
645 
646         cur = self._conn.cursor()
647         if flight_key:
648             cur.execute(
649                 """
650                 SELECT flight_key, timestamp, lat_e6, lon_e6, altitude_m, speed_cms, heading_de2,
651                        vs_cms, squawk, source_ip
652                 FROM trajectory_samples
653                 WHERE flight_key = ? AND timestamp >= ?
654                 ORDER BY timestamp ASC
655                 LIMIT ?
656                 """,
657                 (str(flight_key).strip().upper(), cutoff_iso, limit),
658             )
```